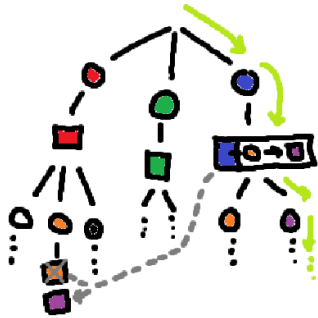
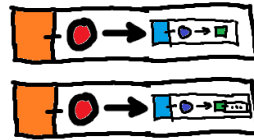


observations: $\circ \text{ (white)} \text{ (red)} \text{ (green)} \text{ (blue)} \text{ (white with } \perp \text{)} \text{ (white with } \top \text{)} \dots \in \mathcal{O}$

actions: $\left\{ \begin{array}{l} \text{orange square, blue square, purple square, black square} \dots \in \mathcal{A}_{-m} \\ \text{orange square} \xrightarrow{\text{red circle}} \text{blue square}, \text{blue square} \xrightarrow{\text{green circle}} \text{purple square} \dots \in \mathcal{A}_m \end{array} \right.$



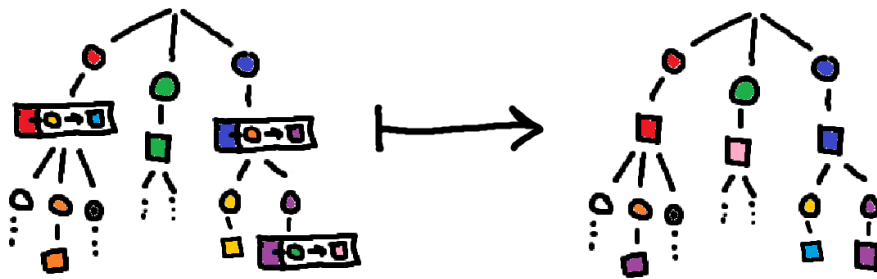
- modification is manipulation
- modifications might chan w/o bound



- modifications are not uniquely optimal and undermine updatelessness
- reflective stability!

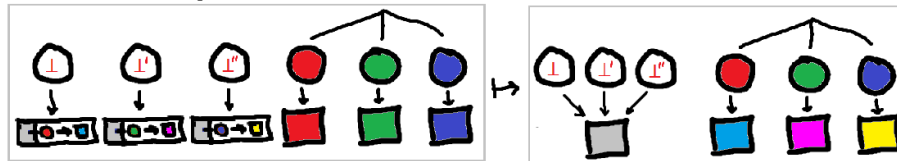
what if we try the obvious unrolling
strategy?

$\forall \boxed{\text{node} \rightarrow \text{node}}, \text{rewrite: } \left\{ \begin{array}{l} \bullet \rightarrow \boxed{\bullet \rightarrow \bullet} \\ \circ \rightarrow \boxed{\circ \rightarrow \bullet} \end{array} \right\} \mapsto \left\{ \begin{array}{l} \bullet \rightarrow \blacksquare \\ \circ \rightarrow \blacksquare \end{array} \right.$
(and forbid nesting)

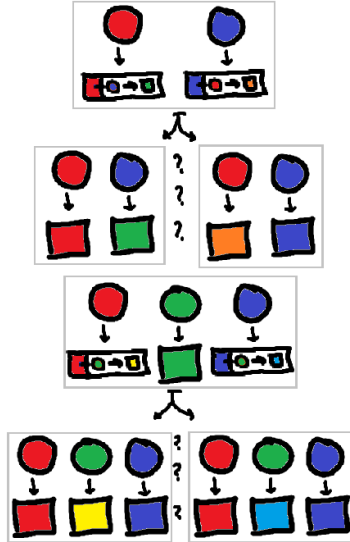


naively unrolling to effective policy fails!

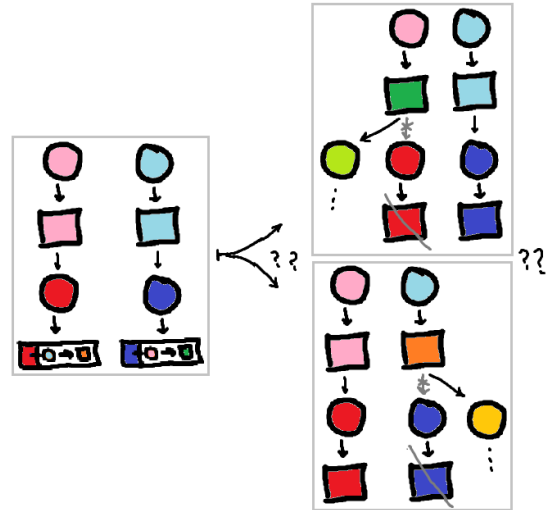
• catastrophic overwriting from impossible counterfactuals



• simple race conditions



• tangled race conditions



Logical Induction + Bayesian LIs

Well-known criterion for reasoning + deciding systems:

"Will it reliably make bad decisions that bankrupt it?"
(Dutch-book / money-pump stance)

e.g. if S has preferences $\text{red} \prec \text{green} \prec \text{blue} \prec \text{red}$ then an adversary

A can offer S : $\left\{ \begin{array}{l} \text{red} + \$1 \rightarrow \text{green} \\ \text{green} + \$1 \rightarrow \text{blue} \\ \text{blue} + \$1 \rightarrow \text{red} \end{array} \right. \Rightarrow S \text{ happily pays A all its money to trade in circles!}$

Logical induction markets (+ Bayesian LIs built on them)
are grounded in this principle as applied to bounded rationality

•

Logical Induction: Basic Idea (Scott G et al)

- * Specify some language $\mathcal{L}$
- * Specify a class of continuous trading strategies, mapping prices to specific trades
- * Specify a class of traders
 - ↳ e.g. poly-time as a function of t , mapping t to a trading strategy
- * Specify initial wealths for traders (eg, $2^{-(\text{description length})}$)
- * Specify schedule for traders to enter the market

LI Basic Idea Continued

Logical Induction Criterion :

- * No trader can exploit the market (unboundedly).
(Basically, money-pump the market.)

Logical Induction Algorithm :

- * At each time t , the "house" sets prices so as to balance traders against each other, based on their wealths. This is possible via Brouwer fixed-point.
 - * Trader wealths at t are updated from $t-1$ based on the new market prices (which are themselves based on wealths), so wealths & prices are one big fixed-point.
-

Prop 1

y	n
---	---

 Prop 2

y	n
---	---

 Prop N

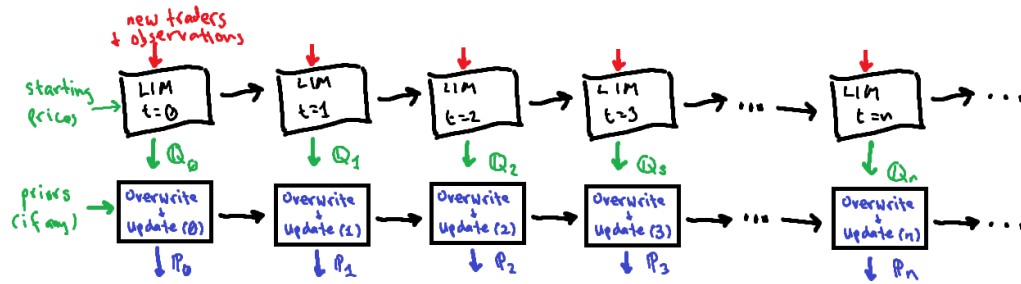
y	n
---	---

Propositional Coherence

- * One way to set up LI is to imagine one "betting table" per proposition. Traders put money on "yes" or "no" indicating their bets, & the house's job is to set odds so that the ratio of money set down = the odds.
- The market will converge towards respecting propositional coherence rules $P(T)=1$, $A \wedge B = 1 \Rightarrow P(A \vee B) = P(A) + P(B)$ because the house rewards arbitrage of failures of these rules.
- Alternatively, we can enforce prop. coh. right away by instead setting up one big betting table on all 2^N boolean combinations.
- Traders bet on a single prop. by spreading money out.
- This only enforces coherence for finitely many prop. at once, but we can set the rest naively assuming independence



Bayesian Logical Inductors



Sam's constraints: Approximation due to some discretization

1. $P_n(\phi) \approx Q_n(\phi)$ for ϕ which are small on day n
2. $P_n(\phi | Q_m = Q_n) := Q[n]$ when $m > n$
3. $P_n(\phi | Q_m = Q_n \wedge Q_{m+1} = Q_{m+1}) = Q_{m+1}$; similarly for longer conjunctions
4. Beliefs $P_{n-1}(Q_n = Q_n)$ are required to balance the equation

Summing over the (finite discretization) $P_{n-1}(\phi) = \sum_{Q_n} P_{n-1}(Q_n = Q_n) Q[\phi]$

Bayesian LI Cont.

to within vanishing ϵ

$\mathbb{P}$ is now a logical inductor, since it agrees with $\mathbb{Q}$ on small prices; furthermore, we can see $\mathbb{P}$'s updates from $\mathbb{P}_n$ to $\mathbb{P}_{n+1}$ as a Bayesian update, since

$$\mathbb{P}_{n+1}(\phi) = \mathbb{P}_n(\phi | Q_{n+1} = Q_{n+1})$$

where Q_{n+1} denotes Q_{n+1} 's actual prices.

Notice that Sam's constraints 1 & 4 imply:

$$\mathbb{P}_{n-1}(\phi) = \sum_{Q_n} \mathbb{P}_{n-1}(Q_n = Q_n) \mathbb{P}_{n-1}(\phi | Q_n = Q_n)$$

This is a version of conservation of expected evidence, AKA no expected net update.

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Bayesian LI Cont.

- + Just listing the constraints doesn't prove that a BLI exists.
- + Constraints 1-3 are easy enough (we can modify large sentences however we like).
- + We can show that #4 is feasible by choosing $P(Q_n = Q_n) = 1$ for one specific Q_n such that $Q_{n-1}(\phi) = Q_n(\phi)$, and $P(Q_n = Q_n) = 0$ otherwise.
- + Obviously this is a very unsatisfying choice!
- + This illustrates how BLI is a very incomplete set of "rationality constraints" for large sentences.

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Two Approaches to Coherently Coherent BLIs

Continuous / real coherence

$$P_t(\phi) = \lim_{\epsilon \rightarrow 0} \int_{Q \in [0,1]^{N_{t+1}}} P_t(\phi | d_{JS}(Q, Q_{t+1}) \leq \epsilon) \cdot P_t(d_{JS}(Q, Q_{t+1}) \leq \epsilon) dQ$$

"how true is ϕ if Q is near Q_{t+1} ?"
 "how likely is it for Q to be that close?"
 d_{JS} : Jensen-Shannon statistical distance

Discrete / bin-grid coherence

$$P_t(\phi) = \lim_{\epsilon \rightarrow 0} \sum_{Q \in [0:\epsilon:1]^{N_{t+1}}} P_t(\phi | \text{bin}(Q) = \text{bin}(Q_{t+1})) \cdot P_t(\text{bin}(Q) = \text{bin}(Q_{t+1}))$$

"how true is ϕ if Q, Q_{t+1} share a bin?"
 "how likely are Q, Q_{t+1} to share the bin?"
 ϵ -bins